Disentangling the Black Box: Interpretable Protein-Ligand Docking via Sparse Autoencoders

Anonymous Author(s)
Affiliation
Address
email

Abstract

Protein-ligand docking pipelines based on variational autoencoders (VAEs) achieve strong predictive performance but produce opaque latent representations, while the large number of candidate poses generated per complex imposes substantial computational cost as each undergoes expensive energy minimization and refinement regardless of quality. Physics-informed generative approaches like the Friesner Group's IFD-ML pipeline ground predictions in thermodynamic principles, making learned features more likely to be physically meaningful and interpretability more tractable and valuable than in end-to-end models. We apply sparse autoencoders (SAEs) to the VAE latent space of the IFD-ML pipeline, training on 435,069 poses across 36 protein systems. At a 4Å threshold, SAE-derived features filter 16.5% of poses prior to refinement while retaining 91.1% of native-like conformations, outperforming the pipeline's own energy score. Fraction-of-variance-explained (FVE) analysis shows the top SAE retains 62% of latent variance while still supporting a high-precision filter. Sparse features activate exclusively on failed docking attempts, each corresponding to a distinct structural failure mode—an interpretable window into what generative models learn about binding physics. These results show SAEs can serve simultaneously as practical computational filters and interpretability tools for building trustworthy drug-discovery pipelines.

1 Introduction

Protein-ligand binding governs essentially all biological processes and underlies the majority of small-molecule drug discovery [Dhakal et al., 2022, Sim et al., 2025]. Predicting how a ligand binds a flexible protein target, the induced fit docking (IFD) problem [Sherman et al., 2006], remains one of computational biochemistry's hardest tasks, addressed by physics-based pipelines such as IFD-MD (IFD with molecular dynamics) [Miller et al., 2021]. Most end-to-end machine learning (ML) approaches, such as DeepDTA and its graph-based successors, treat binding as a sequence-to-structure mapping without explicitly modeling atomic interactions [Wu et al., 2025]; and are further limited by scarce training data, poor interpretability, and weak generalization to novel proteins or ligands [Lam and Katritch, 2025, Jiang et al., 2026].

Schrödinger and the Friesner Group's IFD-ML pipeline combines physics-based simulation with variational autoencoder (VAE) and diffusion generative models to sample conformational space, achieving strong performance in cross-docking benchmarks similar to Mansoor et al. [2024] (Konstantinovsky et al., in preparation). Because it relies on a VAE, each candidate pose is represented by a fixed-dimensional continuous latent vector—the kind of representation a sparse autoencoder (SAE) is designed to decompose into interpretable features (Section 2.2). This creates two problems. First, the pipeline generates thousands of candidate poses per complex, each undergoing expensive Prime energy minimization [Li et al., 2011] and algorithmic refinement regardless of quality—a bottleneck at scale [Wang et al., 2019]. Second, the VAE's latent vectors are opaque: it is unclear which directions encode binding geometry, energetic favorability, or structural quality; and understanding

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why a model scores a pose highly matters as much as the score itself [Qadri et al., 2025]. This paper addresses both: a practical filter that removes likely-poor poses before refinement, and an interpretability method that opens up what the latent space has learned.

Sparse autoencoders (SAEs) decompose opaque neural representations into interpretable, monosemantic features [Templeton et al., 2024, Gao et al., 2024]. Applied to protein language models, SAEs recover biologically meaningful features corresponding to secondary structure, binding sites, and evolutionary conservation [Adams et al., 2025, Simon and Zou, 2025]. Physics-informed pipelines like IFD-ML are a compelling target: because the latent space is constrained by thermodynamic priors and atomic interactions, learned features are more likely to correspond to real physical quantities—energy terms, geometric constraints, binding physics—than in end-to-end models [Li et al., 2026, Wu et al., 2024]. SAEs have been applied to structure-generation models such as RFdiffusion (FoldSAE) [Zarzecki et al., 2025], RFdiffusion3/RoseTTAFold3 [Yu and Olson, 2026], and protein folding representations (ESMFold/ESM2-3B) [Parsan et al., 2025], but to our knowledge not yet to a physics-informed, molecular-docking generative model.

This work applies TopK SAEs to the VAE latent space of the IFD-ML pipeline to serve both goals at once. Concretely, we make three contributions: (1) a practical pre-refinement filter that flags 16.5% of poses for removal while retaining 91.1% of native-like conformations; (2) evidence that the pipeline's own energy score cannot substitute for this filter, structurally unable to reach the SAE's operating points; (3) sparse features that isolate distinct, physically interpretable failure modes, from severe rotational misorientation to milder translational displacement, in what generative models learn about binding physics.

2 Methods

2.1 Dataset

We compiled a dataset of 435,069 molecular docking poses across 36 protein-ligand systems generated by the IFD-ML pipeline, split by case (not pose) to prevent leakage. Each pose consists of a 30-dimensional VAE latent vector, ligand RMSD to the native structure (excluding hydrogen atoms), and a free energy prediction from physics-based scoring. We use two RMSD-based pose-quality definitions: native-like at RMSD $\leq$ 2Å (12.3% native-like, following Shim et al. [2022]), and good at RMSD $\leq$ 4Å (36.6% native-like) for the conservative filtering task. Splits are approximately 70% train, 15% validation, 15% test; normalization uses the training split only, and feature/threshold/combo selection for filtering (Section 2.4) draws on pooled train+val, with test reserved for a single final evaluation.

2.2 TopK Sparse Autoencoder

We train TopK SAEs with $4\times$ expansion (30D input $\rightarrow$ 120D hidden $\rightarrow$ 30D reconstruction) following Adams et al. [2025]; the TopK operation retains the K largest activations and zeros the rest, with reconstruction loss as the training objective. The input is mean-centered via a learned pre-encoder bias, initialized to the training-set mean of the normalized activations and subtracted before encoding, following Simon et al. [2026]'s dead-feature mitigation. We trained SAEs across $K \in \{1, 3, 6, 8, 15, 20, 30\}$, 5 seeds per K, for up to 100 epochs using Adam (lr $= 10^{-3}$, batch size 256), selecting the best run per K by validation loss; we report results for $K \in \{3, 8, 15\}$ in the main text; the full diagnostics sweep across all seven K values is in Appendix A.3 (auPR and FVE trends are consistent, though dead-feature rate is not monotonic in K), selecting the operating K per analysis (Section 3).

2.3 Pose Quality Classification

To assess whether SAE features provide utility beyond raw latents, we trained logistic regression classifiers on (1) 30D normalized VAE latents (L2 penalty, class-balanced) and (2) 120D sparse SAE activations for $K \in \{3, 8, 15, 20, 30\}$ (L1 penalty, class-balanced), fit on train and evaluated on test. auPR is our primary metric, since it handles class imbalance better than ROC-AUC, which we also report for comparison with prior work.

2.4 Threshold-Based Pose Filtering

The paper's central practical result is a pre-refinement filter, built using a three-way train/select/test split so reported performance cannot be inflated by threshold or feature selection on the test set. For each of the 120 features per SAE, we fit an F1-maximizing activation cutoff and rank candidates by precision on a pooled train+val selection set, requiring recall > 0.02 to exclude degenerate near-zero-recall picks. We take the top 16 candidates, then search all subset unions—evaluated on the same selection set—for the union maximizing recall subject to retaining $\geq$80% of native-like poses; the winning union is applied exactly once to test, which is never touched during fitting or selection. To confirm that precision derives from the SAE decomposition rather than raw latent geometry, we apply the identical procedure to 500 random unit directions and all 60 signed raw latent axes. Finally, as a physics-based baseline, we apply the same discipline directly to the IFD-ML pipeline's pre-minimization energy score, accounting for its sentinel value (10,000 kcal/mol)—assigned when the ligand is not in the basin that refinement operates in, or the pose is otherwise too structurally poor to score meaningfully.

3 Results

3.1 Reconstruction, Classification, and Sparsity

Does decomposing the latent space into sparse features cost any of the predictive signal present in the raw, dense latents? Table 1 reports auPR for bad-pose classification, FVE, and dead-feature rate, all on the test set. Raw latents exceed every SAE K on auPR at 4Å and are roughly tied with the best SAE at 2Å (K=3: 0.858 vs. 0.855)—the sparse decomposition provides no consistent predictive lift, so its value lies in filtering (Section 3.2) and interpretability (Section 3.4) rather than classification. FVE grows monotonically with K without a matching auPR gain, and dead-feature rate spikes to 25.8% at K=15 despite the mean-centering mitigation from Section 2.2 [Simon et al., 2026], likely because the fixed 12-slot auxiliary revival budget does not scale with K; stronger, K-scaled interventions may be needed.

Table 1: auPR, FVE, and dead-feature rate, raw latents vs. SAE K (test set).

Rep.	auPR (4Å)	auPR (2Å)	FVE	Dead %
Raw latents	0.770	0.855	—	—
SAE K=3	0.709	0.858	33.7%	2.5%
SAE K=8	0.719	0.821	62.0%	1.7%
SAE K=15	0.750	0.846	90.0%	25.8%

3.2 Practical Filtering

We now evaluate the pre-refinement filter from Section 2.4. Table 2 reports the combo filter for $K \in \{3, 8, 15\}$ at both thresholds, alongside the raw-latent-geometry baselines (Section 2.4). K=8 and K=15 are the reproducible headline operating points, at 4Å and 2Å respectively (K=3's combo cleared the 80% good_kept bar on the selection set at 85.2% but dropped to 77.5% on held-out test—a genuine generalization failure rather than a threshold artifact, and one shared by every $K < 8$ we trained, not just K=3; Appendix A.5): K=8 flags 16.5% of test poses while retaining 91.1% of native-like conformations (precision 0.803); K=15 flags 15.7% while retaining 93.0% (precision 0.935). Recall is modest at both (0.21, 0.17)—a deliberate precision-first design choice, since retaining native-like poses matters more than catching every poor one for a pre-refinement filter.

Neither raw-latent-geometry baseline family reaches the good_kept $\geq$ 80% target at either threshold (0 of 65,535 candidate-union combinations clear it), so Table 2 reports each baseline's closest achievable point instead—well below what K=8/K=15 achieve while flagging fewer poses.

3.3 Comparison to the IFD-ML Pipeline's Energy Score

Applying the same selection discipline (Section 2.4) to the pipeline's pre-minimization energy score, 38.4% of test poses at both thresholds are tied at the sentinel value (energy = 10,000 kcal/mol)—occurring when the ligand starts outside the basin that refinement operates in, or the pose is otherwise unsalvageable. Because this is a tied point mass, no energy threshold can flag fewer than 38.4%

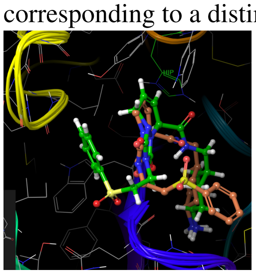
Table 2: Filtering performance: SAE $K \in \{3, 8, 15\}$ vs. raw-latent-geometry baselines, both thresholds, held-out test set.

Threshold	Source	% Flagged	Precision	Recall	Good Kept
4Å	SAE K=3	27.3	0.698	0.301	77.5%
4Å	SAE K=8 (headline)	16.5	0.803	0.210	91.1%
4Å	SAE K=15	12.3	0.879	0.171	95.9%
4Å	Random directions (best)	33.7	0.605	0.321	63.6%
4Å	Raw latent axes (best)	27.8	0.727	0.319	79.2%
2Å	SAE K=3	38.1	0.791	0.353	45.8%
2Å	SAE K=8	19.7	0.884	0.204	84.4%
2Å	SAE K=15 (headline)	15.7	0.935	0.172	93.0%
2Å	Random directions (best)	33.7	0.824	0.325	59.7%
2Å	Raw latent axes (best)	27.8	0.857	0.280	72.8%

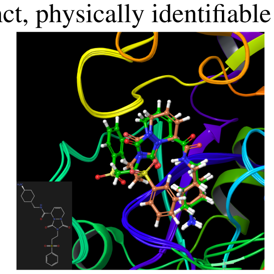
of poses regardless of tuning—both SAE headline operating points fall below this floor and so are structurally unreachable, not merely outperformed. At its most conservative achievable point (the sentinel-only filter, 38.4% flagged), energy reaches 78.9% good_kept at 4Å and 78.2% at 2Å—both well below the SAE filters despite flagging more than double the poses. An unconstrained best-F1 energy threshold performs even worse, degenerating to flagging 100% of poses.

3.4 Interpretable Features

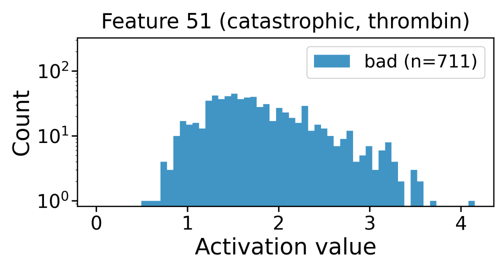
The SAE's sparse features are not just useful for filtering—they are individually interpretable, each corresponding to a distinct, physically identifiable failure mode.



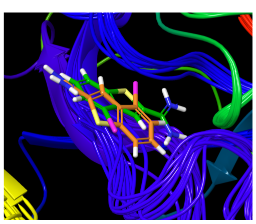
(a) Pose 2384 (RMSD 5.6Å)



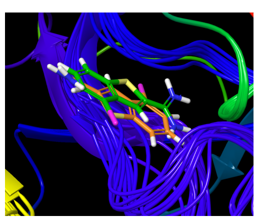
(b) Pose 4556 (RMSD 7.2Å)



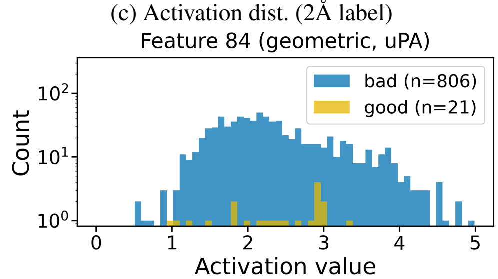
(c) Activation dist. (2Å label)



(d) Pose 11084 (RMSD 5.7Å)



(e) Pose 11377 (RMSD 5.0Å)



(f) Activation dist. (2Å label)

Figure 1: Feature 51 (top, thrombin) and Feature 84 (bottom, uPA), both members of the deployed K=8 4Å combo filter (case details and statistics in text). Left/middle: top-activating test poses, docked pose (orange) vs. reference (green); the colored backbone traces in Feature 51's poses reflect the pipeline's induced-fit receptor-loop sampling, present in both good and bad poses for this case (see below). Right: activation distribution (active poses only, log-scale count) split by the 2Å native-pose criterion.

Table 2's K=8 combo filter is a union of 16 candidate features (Section 2.4); we highlight two with sharply contrasting structural signatures for detailed interpretation. Feature 51 fires on 711 test poses, 99.3% of which come from a single thrombin case pair [Di Cera et al., 1995] (PDB 1AE8→1D9I [De Simone et al., 1997, Krishnan et al., 2000], 706/711; a second thrombin pair accounts for the remaining 5), at precision 0.809 with a sentinel rate of 86.4%. This is not indiscriminate case detection: the feature fires on just 9.5% of that case's poses and 10.8% of its bad poses. Under the stricter, standard 2Å native-pose criterion, precision is perfect: all 711 flagged poses are non-native

(0 false positives). Its highest-activating poses (Figure 1) show a case-specific rotational failure: the ligand's sterically bulky substituent, which should point toward Leu99, instead points toward Ala190—both residues lining the S1/S2 subsites adjacent to the catalytic triad [Bode et al., 1989]. This misorientation is severe enough that most flagged poses fall outside any refinable basin.

Feature 84 tells a different story. It fires on 827 poses, 94.3% concentrated in a single urokinase (uPA) case pair [Spraggon et al., 1995] (PDB 2VIO→1C5W [Frederickson et al., 2008, Katz et al., 2000], 780/827)—the same catalytic serine-protease fold family as thrombin—at precision 0.911 with a sentinel rate of 28.5%, *below* the population baseline; at the 2Å native-pose criterion, precision rises to 0.975 (806/827). Its highest-activating poses show the ligand correctly oriented but translationally displaced within the binding pocket; energies converge to real, physically stable values (clustered around $-9{,}300$ to $-9{,}700$ kcal/mol) rather than the sentinel value. On the 20 most confidently native poses for this case (RMSD $<$0.4Å), feature 84's activation is exactly 0.0000—the feature is silent on good poses.

Together the two features illustrate a rotational-vs-translational split in the geometric errors the SAE decomposes into distinct, monosemantic features: severe rotational misorientation that falls outside any refinable basin (Feature 51), and milder translational displacement that minimization tolerates (Feature 84)—consistent with the monosemanticity hypothesis that sparse features encode distinct failure phenomena [Templeton et al., 2024, Adams et al., 2025, Clark et al., 2026, Bricken et al., 2023].

Case-specific concentration is not unique to Features 51 and 84: across all 16 candidates in the K=8 combo (Table 3, test set), 5 concentrate $\geq$90% of firings in a single case pair and 12 concentrate $\geq$50%, spanning four structurally and functionally distinct targets—the two proteases already discussed, a phosphatase (PTP1B), and an ionotropic glutamate receptor (GluR5)—rather than one fold family. A mirror-image search (features preferring native-like poses) found nothing comparably precise or concentrated.

Table 3: Case-concentration of the K=8 combo filter's 16 candidate features, by dominant protein target (test set).

Target	# features	Mean dom. case %	Max dom. case %
GluR5 (ion channel)	3	81.4%	100.0%
Thrombin (protease)	8	62.2%	99.3%
PTP1B (phosphatase)	1	98.3%	98.3%
uPA (protease)	4	74.2%	95.5%

4 Conclusion

We have shown that sparse autoencoders can decompose the latent space of a physics-informed docking pipeline into interpretable, case-specific features—high-precision detectors for distinct, rare failure modes. The deployed combo filters remove a substantial share of poor poses while retaining the large majority of native-like conformations at both thresholds (Section 3.2), beating both a raw-latent-geometry baseline and the pipeline's own energy score, structurally unable to reach these operating points (Section 3.3). SAEs offer a path toward docking pipelines that are not only predictive but mechanistically understood—systems whose failures can be diagnosed, explained, and, eventually, corrected.

That case-specific concentration extends across the K=8 combo's 16 features (Section 3.4) suggests the SAE learns protein-specific failure signatures a global classifier would dilute—these filters are most valuable when the target family is known, complementary to case-agnostic scoring. Recall is modest at both headline operating points, a deliberate precision-first choice rather than a detectability ceiling, and generalization beyond the 36 systems studied here requires further validation. Understanding a generative pipeline's failures is one step closer to making physics-informed drug discovery interpretable.

Broader Impact This work reduces computational cost and improves interpretability by filtering low-quality poses before refinement and surfacing failure modes the model has not learned. Filtering existing poses, rather than generating new molecules, does not lower the barrier to designing toxic compounds—the field's primary dual-use concern. A more realistic risk is uneven performance: our filters are case-specific (Section 3.4), so precision may not transfer beyond these 36 systems, and over-reliance could deprioritize rarer disease targets.

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A Appendix

A.1 Full case-concentration breakdown for the K=8 combo filter

Table 4 extends Table 3 to all 16 individual candidate features (test set), rather than aggregating by protein target. All 16 are members of the deployed K=8 combo (Section 3.2).

Table 4: Per-feature case-concentration for all 16 candidates in the K=8 4Å combo filter (test set), sorted by dominant-case %.

Feature	n firing	Precision	Dominant case	Dom. %	# cases
43	196	0.964	GluR5 (3fv1→3fuz)	100.0	1
51	711	0.809	Thrombin (1ae8→1d9i)	99.3	2
98	482	0.263	PTP1B (1c84→2h4g)	98.3	2
2	841	0.735	uPA (2vio→1c5w)	95.5	3
84	827	0.911	uPA (2vio→1c5w)	94.3	3
91	587	0.876	Thrombin (1gj5→1d3d)	83.5	5
100	958	0.902	Thrombin (1gj5→1d3d)	80.2	3
28	25	0.720	GluR5 (3fv1→3fuz)	76.0	2
102	493	0.813	GluR5 (3fv1→3fuz)	68.2	4
53	179	0.480	uPA (2vio→1c5w)	64.8	3
10	594	0.827	Thrombin (1gj5→1d3d)	56.6	5
85	3709	0.942	Thrombin (1ae8→1d9i)	52.1	5
71	1718	0.574	Thrombin (1ae8→1d9i)	49.5	5
12	446	0.863	uPA (2vio→1c5w)	42.2	5
50	360	0.742	Thrombin (1ae8→1d9i)	41.9	5
118	78	0.500	Thrombin (1ae8→1d9i)	34.6	6

A.2 Mirror-image feature search

Section 3.4 reports that a search for the mirror-image case—features that prefer native-like poses rather than flagging bad ones—found nothing comparably precise or concentrated. The best such feature reached only 55.3% precision at 1.6% recall (4Å label) and 22.2% precision at 11.1% recall (2Å label), below all but the three lowest-precision features in Table 4. This asymmetry is consistent with the SAE's TopK sparsity favoring high-precision detectors for rare, structurally distinctive failure modes over the more diffuse signal of "everything went right."

A.3 Full K sweep diagnostics

Table 5 extends Table 1 to all seven trained K values. Dead-feature rate is not monotonic in K: it is worst at K=1 (80.8%, where the single-active-feature constraint likely dominates), falls to near zero for $K \in \{3, 6, 8\}$, rises again for $K \in \{15, 20\}$, and partially recovers at K=30. The fixed-auxk explanation offered in Section 3.1 for the K=3/8-vs-15 comparison is therefore only a partial account of this pattern, not a complete one—the full sweep suggests at least two distinct regimes (extreme-sparsity starvation at very low K, and revival-budget saturation at high K) rather than a single monotonic cause.

Table 5: auPR, FVE, and dead-feature rate across the full trained K sweep (test set).

K	auPR (4Å)	auPR (2Å)	FVE	Dead %
1	0.660	0.838	−0.2%	80.8%
3	0.709	0.858	33.7%	2.5%
6	0.737	0.867	49.5%	0.8%
8	0.719	0.821	62.0%	1.7%
15	0.750	0.846	90.0%	25.8%
20	0.749	0.846	98.0%	30.8%
30	0.742	0.810	100.0%	16.7%

A.4 Case-concentration for the K=15 2Å combo

Table 6 reports the same case-concentration analysis as Table 4, but for the K=15 2Å headline combo's 4 features rather than K=8's 16. This combo is less case-locked overall—two of its four features spread across 5–6 distinct cases rather than concentrating like most of K=8's candidates—except for Feature 119, which is both small (n=76) and perfectly case-specific. Feature 81 is nominally part of the combo but flags zero test poses (Section 3.1), consistent with it being dead on test.

Table 6: Case-concentration for the K=15 2Å combo filter's 4 features (test set).

Feature	n firing	Precision	Dominant case	Dom. %	# cases
119	76	1.000	Thrombin (1ae8→1d9i)	100.0	1
101	9235	0.948	GluR5 (3fv1→3fuz)	58.6	6
23	1472	0.849	uPA (2vio→1c5w)	53.6	5
81	0	—	(dead on test)	—	0

A.5 Full K sweep: does the selection-set operating point generalize to test?

Section 3.2 shows K=3's combo clears the 80% good_kept bar on the selection set (85.2%) but fails on held-out test (77.5%). Table 7 repeats this selection-vs-test comparison for the full K sweep at 4Å. Every combo clears the bar on the selection set (by construction), but only $K \geq 8$ reliably holds up on test—$K \in \{1, 3, 6\}$ all drop below 80% on test, with K=6 borderline (79.9%). This is a systematic pattern, not an isolated failure at K=3: the selection procedure generalizes reliably once K is large enough, and K=8 is the smallest K at which it consistently does.

Table 7: Selection-set vs. test-set good_kept for the winning combo at each trained K (4Å).

K	Selection good_kept	Test good_kept	Test precision	Test recall
1	82.4%	75.6%	0.575	0.191
3	85.2%	77.5%	0.698	0.301
6	82.8%	79.9%	0.754	0.355
8	83.6%	91.1%	0.803	0.210
15	80.8%	95.9%	0.879	0.171
20	84.9%	86.1%	0.621	0.132
30	82.2%	94.7%	0.817	0.137

A.6 Full protein system list

The dataset spans 36 case pairs across 23 distinct protein targets (some targets contribute multiple case pairs, e.g. thrombin, PTP1B, PIM1). Each case ID follows the pattern `{target}_{PDB_A}_{target}_{PDB_B}`, identifying a cross-docking pair between the two named structures (the same PDB pair appears in Section 3.4 as "PDB A→B"). Two IDs deviate from this pattern and are reproduced verbatim from the pipeline's case labels rather than corrected: `dpp4_1n1m_HIP740_dpp4_2fjp` carries an extra ligand/residue identifier (`HIP740`), and `dhodh_4jgd_dhodh_3u2`'s second code is 3 characters rather than the standard 4-character PDB ID.

The full list of case pairs: afab_2nnq_afab_3fr4, ask1_4bhn_ask1_4bie, bace1_5clm_bace1_5ezx, cdk2_1g5s_cdk2_2b55, cdk2_1pxj_cdk2_2btr, chk1_2brb_chk1_2hxl, chk1_2c3k_chk1_2cgu, chk1_2c3k_chk1_2cgx, chk1_2cgu_chk1_2ayp, dhodh_4jgd_dhodh_3u2, dpp4_1n1m_HIP740_dpp4_2fjp, dpp4_1n1m_dpp4_2oph, fxa_1wu1_fxa_1xkb, glur5_3fv1_glur5_3fuz, gsk3b_3sd0_gsk3b_4ach, hivrt_2b5j_hivrt_1c0u, hsd1_2irw_hsd1_3ch6, hsp90_1yet_hsp90_2byi, mdm2_3jzk_mdm2_4zyf, pde5_4md6_pde5_3shy, pdk_3sc1_pdk_3nun, pim1_2xiz_pim1_4lmu, pim1_2xj2_pim1_4enx, pim1_3vbt_pim1_4lmu, pim1_4lmu_pim1_4bzo, ptp1b_1c84_ptp1b_1xbo, ptp1b_1c84_ptp1b_2h4g, ptp1b_1no6_ptp1b_1pyn, rho_2esm_rho_2etk, rock1_4yvc_rock1_5wne, throm_1ae8_throm_1d9i, throm_1gj5_throm_1d3d, throm_1k21_throm_1d3d, tryp_2bm2_tryp_3v7t, tryp_5f03_tryp_2za5, upa_2vio_upa_1c5w. The 6 test-set systems (Section 3.4) are throm_1ae8_throm_1d9i, throm_1gj5_throm_1d3d, glur5_3fv1_glur5_3fuz, upa_2vio_upa_1c5w, ptp1b_1c84_ptp1b_2h4g, and hsd1_2irw_hsd1_3ch6.

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